

## LACON, IL, USA

WMO: 720141

Lat: 41.019N

Lon: 89.386W

Elev: 173

StdP: 99.26

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04868

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 1             | -19.3      | -16.4 | -23.5                         | 0.5 | -18.4 | -21.2 | 0.6 | -15.7 | 10.4                  | -1.0 | 9.1 | -1.8 | 2.0                      | 310  | 0.471 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.7                   | 33.0            | 24.0 | 32.1 | 23.8 | 30.7 | 23.1 | 26.1                | 31.0 | 25.1 | 29.7 | 24.2 | 28.6 | 3.3                  | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.8                            | 20.3 | 29.1 | 23.8 | 19.0 | 27.8 | 22.8 | 17.9 | 27.0 | 82.0          | 30.8 | 77.4 | 29.6 | 73.7 | 28.4 | 29.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 9.0               | 8.0  | 7.2 | -23.8                      | 35.4 | 4.5                | 2.0 | -27.0                                              | 36.8 | -29.6      | 38.0 | -32.1      | 39.2 | -35.4      | 40.7 |  |  |
|                   |      |     | -24.1                      | 28.0 | 4.3                | 0.9 | -27.2                                              | 28.6 | -29.8      | 29.1 | -32.2      | 29.6 | -35.3      | 30.2 |  |  |
|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
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|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
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|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
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|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 11.0  | -4.0  | -2.7  | 4.7   | 10.9  | 17.1  | 22.0  | 23.7  | 22.8  | 19.8  | 12.3  | 5.5   | -1.0  | (6)  |
| (7)  |                                                                                  | DBStd       | 11.15 | 6.99  | 6.79  | 6.28  | 5.26  | 4.82  | 3.33  | 3.33  | 3.02  | 4.11  | 5.40  | 5.60  | 6.08  | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1570  | 435   | 356   | 189   | 51    | 4     | 0     | 0     | 0     | 1     | 36    | 156   | 342   | (8)  |
| (9)  |                                                                                  | HDD18.3     | 3239  | 693   | 588   | 425   | 227   | 83    | 6     | 2     | 3     | 30    | 199   | 386   | 598   | (9)  |
| (10) |                                                                                  | CDD10.0     | 1937  | 1     | 2     | 24    | 78    | 223   | 361   | 424   | 398   | 295   | 109   | 20    | 2     | (10) |
| (11) |                                                                                  | CDD18.3     | 565   | 0     | 0     | 2     | 5     | 44    | 117   | 167   | 142   | 75    | 13    | 0     | 0     | (11) |
| (12) |                                                                                  | CDH23.3     | 5230  | 0     | 0     | 12    | 73    | 411   | 1054  | 1622  | 1254  | 691   | 112   | 1     | 1     | (12) |
| (13) |                                                                                  | CDH26.7     | 1753  | 0     | 0     | 1     | 8     | 109   | 342   | 615   | 425   | 230   | 24    | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.9   | 3.4   | 3.4   | 3.5   | 3.7   | 3.0   | 2.5   | 1.9   | 1.8   | 2.1   | 2.7   | 3.2   | 3.1   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 909   | 52    | 45    | 65    | 86    | 108   | 100   | 98    | 97    | 71    | 76    | 72    | 47    | (15) |
| (16) |                                                                                  | PrecMax     | 1250  | 111   | 148   | 140   | 128   | 237   | 258   | 269   | 191   | 137   | 197   | 144   | 80    | (16) |
| (17) |                                                                                  | PrecMin     | 553   | 8     | 13    | 25    | 42    | 13    | 27    | 9     | 28    | 15    | 20    | 2     | 24    | (17) |
| (18) |                                                                                  | PrecStd     | 149   | 28    | 33    | 36    | 31    | 63    | 56    | 58    | 48    | 36    | 45    | 39    | 18    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 14.8  | 17.6  | 25.1  | 27.7  | 32.3  | 34.1  | 37.2  | 34.9  | 33.7  | 29.8  | 22.3  | 17.8  | (19) |
| (20) |                                                                                  |             | MCWB  | 11.8  | 12.3  | 17.1  | 16.3  | 22.9  | 24.4  | 25.0  | 23.6  | 23.5  | 20.9  | 14.6  | 14.4  | (20) |
| (21) |                                                                                  | 2%          | DB    | 11.0  | 13.1  | 21.0  | 25.6  | 29.4  | 32.2  | 33.4  | 32.4  | 31.9  | 26.3  | 18.8  | 13.0  | (21) |
| (22) |                                                                                  |             | MCWB  | 8.6   | 10.1  | 14.8  | 16.5  | 20.9  | 23.5  | 24.8  | 23.9  | 22.6  | 18.2  | 14.2  | 10.2  | (22) |
| (23) |                                                                                  | 5%          | DB    | 7.6   | 9.8   | 17.5  | 22.9  | 27.5  | 30.7  | 32.1  | 31.0  | 29.2  | 23.9  | 16.4  | 10.3  | (23) |
| (24) |                                                                                  |             | MCWB  | 5.8   | 7.0   | 13.2  | 14.8  | 19.2  | 22.5  | 24.5  | 23.5  | 21.7  | 17.7  | 12.3  | 8.3   | (24) |
| (25) |                                                                                  | 10%         | DB    | 4.8   | 6.2   | 14.7  | 21.0  | 25.8  | 28.4  | 30.1  | 28.9  | 27.4  | 21.3  | 14.0  | 7.6   | (25) |
| (26) |                                                                                  |             | MCWB  | 3.0   | 3.9   | 10.7  | 14.0  | 18.5  | 21.6  | 23.6  | 22.3  | 19.7  | 16.4  | 10.5  | 5.9   | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 13.2  | 14.3  | 18.3  | 19.6  | 24.0  | 26.4  | 27.7  | 27.1  | 25.2  | 22.6  | 16.9  | 15.2  | (27) |
| (28) |                                                                                  |             | MCD B | 14.7  | 16.3  | 23.9  | 24.6  | 30.2  | 31.7  | 33.1  | 31.8  | 31.2  | 27.8  | 20.1  | 17.2  | (28) |
| (29) |                                                                                  | 2%          | WB    | 9.1   | 10.0  | 15.8  | 17.7  | 22.4  | 25.0  | 26.5  | 25.6  | 24.0  | 20.7  | 14.8  | 11.3  | (29) |
| (30) |                                                                                  |             | MCD B | 10.4  | 12.9  | 19.6  | 23.3  | 27.4  | 29.3  | 31.2  | 30.5  | 29.4  | 24.0  | 17.6  | 12.5  | (30) |
| (31) |                                                                                  | 5%          | WB    | 5.8   | 6.8   | 13.9  | 16.3  | 21.0  | 23.9  | 25.5  | 24.5  | 22.8  | 19.2  | 12.9  | 8.2   | (31) |
| (32) |                                                                                  |             | MCD B | 7.7   | 9.4   | 16.8  | 21.3  | 25.3  | 28.4  | 29.9  | 28.7  | 27.4  | 22.1  | 15.9  | 9.8   | (32) |
| (33) |                                                                                  | 10%         | WB    | 3.1   | 4.0   | 10.8  | 14.7  | 19.7  | 22.9  | 24.5  | 23.5  | 21.7  | 17.2  | 11.0  | 5.9   | (33) |
| (34) |                                                                                  |             | MCD B | 4.6   | 6.2   | 14.3  | 19.9  | 24.3  | 27.3  | 28.9  | 27.3  | 25.6  | 20.8  | 14.0  | 7.6   | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        |       | 8.4   | 8.9   | 10.1  | 11.8  | 11.5  | 10.9  | 10.7  | 11.0  | 12.2  | 11.2  | 9.4   | 7.9   | (35) |
| (36) |                                                                                  | 5% DB       | MCDBR | 10.8  | 13.0  | 13.1  | 14.9  | 13.4  | 12.4  | 11.9  | 12.6  | 13.2  | 14.1  | 12.6  | 10.9  | (36) |
| (37) |                                                                                  |             | MCWBR | 8.8   | 9.8   | 8.6   | 8.3   | 6.8   | 6.0   | 5.5   | 5.7   | 5.9   | 7.6   | 8.5   | 9.1   | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 10.4  | 12.1  | 12.0  | 12.9  | 11.8  | 10.4  | 10.6  | 10.1  | 11.2  | 10.8  | 11.0  | 10.4  | (38) |
| (39) |                                                                                  |             | MCWBR | 8.9   | 9.7   | 8.6   | 8.3   | 6.7   | 5.7   | 5.7   | 5.6   | 5.6   | 7.4   | 8.2   | 9.4   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |       | 0.292 | 0.309 | 0.342 | 0.393 | 0.427 | 0.437 | 0.453 | 0.437 | 0.404 | 0.350 | 0.308 | 0.291 | (40) |
| (41) |                                                                                  | taud        |       | 2.438 | 2.388 | 2.396 | 2.258 | 2.221 | 2.238 | 2.177 | 2.218 | 2.281 | 2.436 | 2.502 | 2.455 | (41) |
| (42) |                                                                                  | Ebn at Noon |       | 869   | 903   | 909   | 878   | 853   | 842   | 823   | 826   | 830   | 845   | 845   | 839   | (42) |
| (43) |                                                                                  | Edh at Noon |       | 78    | 95    | 107   | 131   | 140   | 138   | 145   | 135   | 119   | 91    | 73    | 71    | (43) |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      |       | 1.77  | 2.69  | 3.68  | 4.72  | 5.45  | 6.19  | 6.17  | 5.43  | 4.60  | 3.08  | 2.03  | 1.49  | (44) |
| (45) |                                                                                  | RadStd      |       | 0.20  | 0.29  | 0.29  | 0.40  | 0.40  | 0.47  | 0.35  | 0.34  | 0.38  | 0.36  | 0.17  | 0.19  | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (52 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page